Polyphenols and antioxidant properties of almond skins: influence of industrial processing.

Almond (Prunus dulcis[Mill.] D.A. Webb) skins have been proposed as a source of bioactive polyphenols. In this article, the phenolic composition and antioxidant activity of almond skins obtained from different processes (blanching [freeze-drying], blanching + drying, and roasting) were studied. A total of 31 phenolic compounds corresponding to flavan-3-ols (33% to 56% of the total of phenolic compounds identified), flavonol glycosides (9% to 36%), hydroxybenzoic acids and aldehydes (6% to 26%), flavonol aglycones (1.7% to 18%), flavanone glycosides (3% to 7.7%), flavanone aglycones (0.69% to 5.4%), hydroxycinnamic acids (0.65% to 2.6%), and dihydroflavonol aglycones (0% to 2.8%) were determined in the skins from 3 different varieties of almonds. The total contents of phenolic compounds identified were significantly ($P < 0.05$) higher (around 2-fold) in the roasted samples than in the blanched almonds (freeze-dried). Industrial drying (oven drying) of the blanched almond skins produced an increase (< 2-fold) in the contents of phenolic compounds, although the results were only statistically significant ($P < 0.05$) for some samples. The antioxidant activity (ORAC values) was higher for the roasted samples (0.803 to 1.08 mmol Trolox/g), followed by the samples subjected to blanching + drying (0.398 to 0.575 mmol Trolox/g) and then the blanched (freeze-dried) samples (0.331 to 0.451 mmol Trolox/g). Roasting is the most suitable type of industrial processing of almonds to obtain almond skin extracts with the greatest antioxidant capacity.